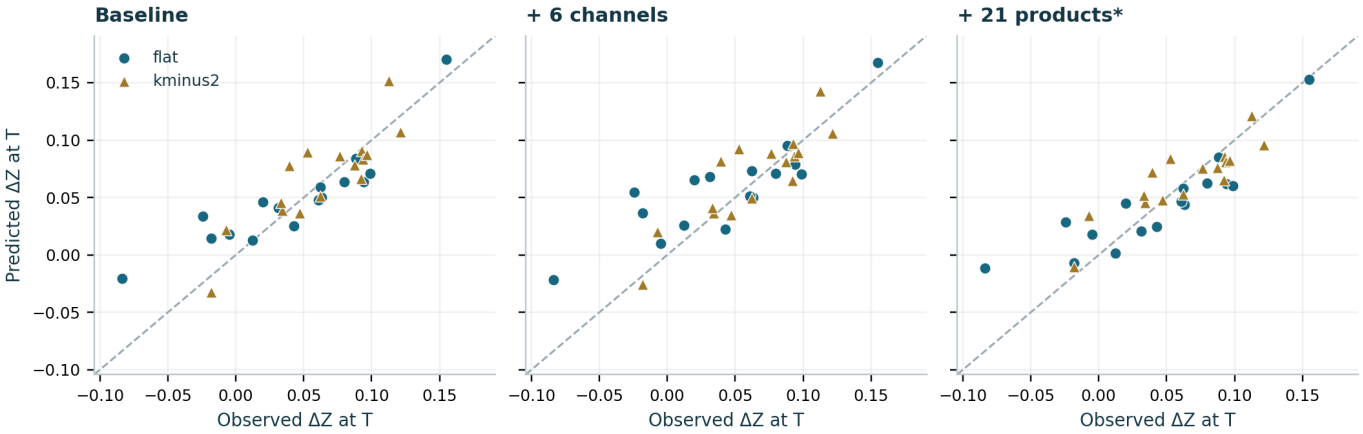
# Phase organization in a controlled forecast test

## No held-out gain from the primary organization model

Across 32 held-out phase realizations, adding the six initial stretching channels to the spectrum-plus-current-stretching baseline did not improve the primary forecast. Pooled MSE increased 30.62%. The paired 95% interval for baseline-minus-augmented MSE is [-0.000449, +1.301e-05]. It spans zero, so the sign of the average gain remains uncertain in this small test sample.

Primary endpoint: ΔZ = Z(0.5)-Z(0) | 64 training / 32 held-out cases




Each point is one untouched test case. The dashed diagonal is perfect prediction. Identical axes preserve comparison. *Products: secondary quadratic amendment.

| Model | α | MAE | RMSE | MSE |
|---|---|---|---|---|
| Baseline | 10 | 0.0197681 | 0.0248366 | 0.000616859 |
| + 6 channels | 10 | 0.0216931 | 0.0283853 | 0.000805725 |
| + 21 products* | 100 | 0.0192707 | 0.024851 | 0.000617571 |
| Fixed Euler | fixed | 0.254508 | 0.277304 | 0.0768975 |

Primary comparison: Baseline versus + 6 channels. Smaller errors are better. Euler is a fixed initial-rate extrapolation, included as a reference.

## A secondary check does not change the primary result

The separately frozen quadratic amendment added all 21 channel pair products. Its pooled MSE increased 0.12% relative to baseline; the MSE difference is -7.122e-07, with 95% interval [-0.0001568, +0.0001642]. This is a secondary representation check, not a replacement for the original primary test. Its interval is unadjusted for exploratory multiplicity.

## The useful distinction

Phase changes the interaction structure while the initial magnitude inventory stays fixed. That is an exact design property. Useful extra prediction from this compact channel representation is a separate claim, and it was not established here.

# A pure phase-only experiment

Breath, sleep transitions, musical phrasing, DNA helices, and attunement motivated the question: can organization change an outcome when an inventory of constituents is held fixed? Here that idea becomes a narrow fluid experiment: preserve every initial Fourier magnitude and polarization, vary common spatial phases, and test a future outcome. No physiological mechanism or intervention is represented, and a Fourier curl eigenpolarization is not a DNA strand.

The inventory is a set of summaries, not the full initial velocity field. A fully specified smooth velocity, viscosity, forcing, and boundary conditions determine the local classical evolution. A change in trajectories under phase variation therefore does not imply indeterminacy of Navier-Stokes.

## A pure common-phase ensemble

The 96 cases use 2 spectral profiles, 64 training cases, and 32 untouched test cases (flat: 32 train / 16 test, kminus2: 32 train / 16 test). All production runs use N=24, $\nu=0.0357143$, $\Delta t=0.005$, $T=0.5$, on the periodic cube $[0,2\pi]^3$. The flat profile uses constant modal amplitude; kminus2 uses amplitude $|k|^{-2}$, hence power proportional to $|k|^{-4}$.

$c_{\pm}(k) = A_k \exp[i(\theta_k \pm \delta_k)]$ ; $\delta_k$ fixed, $\theta_k$ independently varied

Each velocity coefficient is a common phase times a fixed real transverse vector. The conjugate partner enforces reality. This fixes every Cartesian and helical modal power, not only the total spectrum. A spatial translation cancels in each closed triad phase sum.

| Profile | E(0) | Z(0) | D(0) | H(0) |
|---|---|---|---|---|
| flat | 1 | 2.63636 | 18.7273 | 0 (roundoff) |
| kminus2 | 1 | 1.40231 | 5.19958 | 0 (roundoff) |

$E=\langle|u|^2\rangle/2$, $Z=\langle|\omega|^2\rangle/2$, $D=\langle|\nabla\omega|^2\rangle$, $H=\langle u\cdot\omega\rangle$, $S=\langle\omega\cdot[(\omega\cdot\nabla)u]\rangle$. Thus $Z'=S-\nu D$. The values above are reference inputs; H=0 is an initial control, not a claim that viscous evolution preserves zero H.

| Within-profile control | flat | kminus2 |
|---|---|---|
| Cartesian modal power | 8.33e-17 | 1.53e-16 |
| Helical modal power | 4.86e-17 | 1.11e-16 |
| Energy E | 1.11e-16 | 2.22e-16 |
| Enstrophy Z | 8.88e-16 | 4.44e-16 |
| Dissipation measure D | 7.11e-15 | 1.78e-15 |
| Helicity H | 3.80e-16 | 2.89e-16 |

Maximum absolute differences from each profile reference, extracted from saved initial fields. All are at floating-point roundoff.

## Full resolved stretching, including mixed bands

Split each helical sign into low/high radial bands at $|k|=1.5$. The four real component fields generate $4^3=64$ ordered blocks $T_{abc}=\langle\mathrm{curl}(u_a)\cdot[(\mathrm{curl}(u_b)\cdot\nabla)u_c]\rangle$. Their sum is full resolved S. No retained high mode or mixed interaction is omitted. These are stretching blocks, not spectral energy fluxes.

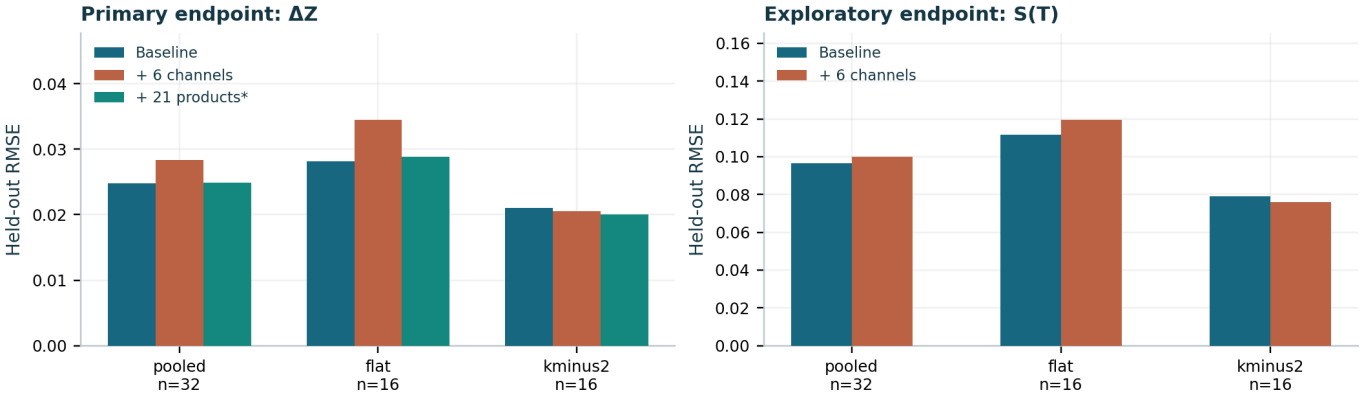
Grouping by the number of positive-helicity legs, or by the number of high-band legs, gives two four-channel sums equal to S. The model retains counts 0, 1, 2 from each grouping. The complete blocks and both count-3 channels remain in the data.

The six supplied channels are not six independent dimensions. This initial support has no all-high triads: max $|n_{high3}|=2.518e-16$, and max $|n_{high0}+n_{high1}+n_{high2}-S|=1.332e-15$. Redundant predictors change ridge's penalty geometry. A gain, had it occurred, would compare model representations; it would not prove that a new independent physical quantity is necessary or sufficient.

# What the forecast actually tested

Baseline: one unpenalized intercept and a separate cubic in S(0) for each profile. Within this ensemble, profile identity encodes the entire fixed magnitude spectrum and E, Z, D, H. Primary augmentation: the same baseline plus nplus0..2 and nhigh0..2. All changing predictors use t=0 only. The fixed Euler reference is T[S(0)-νD(0)]; it is not the main learned comparator.

Four deterministic, profile-stratified training folds select ridge α from 0.0001, 0.01, 0.1, 1, 10, 100. Each fold fits its own means and standard deviations; final fits use all training data only. The untouched test targets are excluded from preprocessing, feature selection, tuning, and model selection. The primary plan and separate pre-outcome quadratic amendment are preserved with SHA256 checks.




Errors differ by profile. *The product representation is exploratory. Euler is omitted from this chart for scale and remains fully reported on page 1 and in the data.

| Target / model | Pooled MSE | flat MSE | kminus2 MSE |
|---|---|---|---|
| ΔZ: Baseline | 0.00061686 | 0.00079157 | 0.00044215 |
| ΔZ: + 6 channels | 0.00080572 | 0.0011894 | 0.00042208 |
| ΔZ: + 21 products* | 0.00061757 | 0.00083414 | 0.00040101 |
| S(T): Baseline | 0.0093487 | 0.012438 | 0.006259 |
| S(T): + 6 channels | 0.010016 | 0.014283 | 0.0057496 |

| Target / model | α | Training CV MSE | Design rank |
|---|---|---|---|
| ΔZ: Baseline | 10 | 0.001366 | 6 / 6 |
| ΔZ: + 6 channels | 10 | 0.0014828 | 11 / 12 |
| ΔZ: + 21 products* | 100 | 0.0018034 | 31 / 33 |
| S(T): Baseline | 10 | 0.020922 | 6 / 6 |
| S(T): + 6 channels | 100 | 0.021037 | 11 / 12 |

Ranks apply to standardized penalized designs after removing profile intercepts. The nominal feature count exceeds rank; positive ridge penalties keep the fits well-defined.

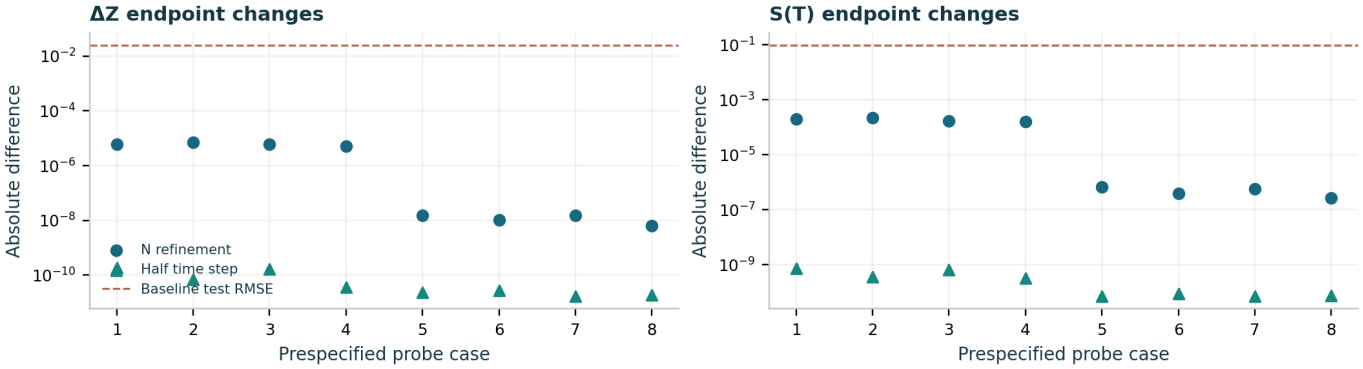
## Uncertainty is narrow in scope

Intervals use 2,000 paired bootstrap resamples of held-out errors, independently within each profile and preserving its size. They condition on the fitted models, omit training-sample variability and discretization uncertainty, and describe only independent draws from this synthetic phase distribution. Per-profile samples are small; pooled results can conceal heterogeneous errors.

Exploratory S(T): baseline-minus-augmented MSE -0.0006673; 95% interval [-0.001855, +0.0004468]. The planned 100-shuffle diagnostic gives an add-one exceedance fraction of 0.7921. Whole organization vectors were shuffled within profile and split. This breaks their dependence on S, so the fraction is descriptive only, not a calibrated conditional-randomization p-value.

# Numerical checks and limits

| Refinement | Cases | Max field L2 | Max \|δΔZ\| | Max \|δS(T)\| |
|---|---|---|---|---|
| N 24 → 32 | 8 | 1.156e-03 | 7.189e-06 | 2.175e-04 |
| Δt 0.005 → 0.0025 | 8 | 6.848e-10 | 1.872e-10 | 7.397e-10 |




The table and plot compare the same saved initial fields against independent spatial and time-step refinements. Probes were specified before outcomes and never replace production targets. The observed endpoint changes are much smaller than held-out prediction errors in these sampled cases. They support numerical adequacy for this comparison; they do not certify all cases or a continuum limit.

Production maxima: divergence L2 7.2e-16; helical / band closure discrepancies 1.305e-15 / 1.346e-15; outer-shell enstrophy fraction 0.000125; reported CFL 0.08469. Largest sampled energy increase: 0. RHS balance residuals check semidiscrete algebra, not integrated time-stepping accuracy.

Independent review reproduced all fitted predictions, CV scores, paired bootstrap draws, and organization shuffles, and checked targets against saved fields. See phase2/INDEPENDENT_REVIEW.md.

## Navier-Stokes remains the primary mathematical question

Only one viscosity, one short horizon, a sparse prescribed mode set, fixed polarizations, and two magnitude profiles are tested. The features compress signed interactions and may discard useful organization. Failure of these models does not show that phase lacks information; it limits the tested representation and fitting procedure. No new features or model changes were selected after seeing the held-out outcomes.

Nothing here proves global regularity, blowup, universal closure, causal sufficiency, or an ordering principle valid for arbitrary flows. The unresolved analytical target remains an a priori, cutoff-independent bound on cumulative normalized stretching excess, valid for every time prefix. A finite-horizon prediction score cannot supply that bound. Phase organization is physically relevant to interactions, but this test did not establish a useful compact predictive improvement beyond its declared baseline.

## Reproduce and inspect

From helical_lab: python -m phase2.make_report

The report generator reads saved results, verifies provenance, and independently recomputes held-out metrics. To regenerate the experiment: run phase2.ensemble prepare, run, and probes; then phase2.predict and phase2.review_phase2. See REPORT.md for complete commands, checksums, and source locations.

Complete initial/final fields and numerical checks: results/phase2/ensemble/. Predictions, models, CV grids, paired bootstrap samples, and every shuffle: results/phase2/prediction/. Mathematical definitions: CONTROLLED_ENSEMBLE_MATH.md and DERIVATION.md. Prospective decisions: PREDICTION_PLAN.md and PREDICTION_AMENDMENT.md.

Background: Waleffe (1991), helical decomposition and nonlinear interactions. Bundled fonts: DejaVu Sans; license in assets/fonts/LICENSE.txt.

Prediction plan SHA256: 02162dba49bf33d2986ffbf4eeb9745de4995ebf3498f78b8f1ded7e78671f6d

SEPARATE CALCULATION / POST-EXPERIMENT EXACT CERTIFICATE

# Exact closure obstruction

## A local obstruction to exact closure by magnitudes and initial stretching

This is a separate, post-experiment exact calculation outside the prediction ensemble. It leaves the negative primary forecast result unchanged. Its question is whether complete initial Fourier magnitudes and current stretching determine the local future enstrophy.

$u_s(x) = \Sigma(j=1..3)\ q_j \cos(k_j \cdot x) + s\ q_4 \cos(k_4 \cdot x),\ s = \pm 1$

| Mode j | Wavevector k_j | Polarization q_j |
|---|---|---|
| 1 | (1, 0, 0) | (0, 1, -1) |
| 2 | (1, 1, 0) | (-1, 1, 0) |
| 3 | (0, 1, 1) | (-1, -1, 1) |
| 4 | (2, 2, 1) | (-2, 1, 2) |

Each $k_j \cdot q_j = 0$. Both fields are smooth, real, mean-zero and divergence-free. The sign flip preserves every Cartesian and helical Fourier magnitude. Parity gives $H = S = 0$. Common initial values:

| E | Z | D | H | S |
|---|---|---|---|---|
| 4 | 93/4 | 751/2 | 0 | 0 |

## Same initial slope, different exact curvature

| Exact initial quantity | s = +1 | s = -1 |
|---|---|---|
| $Z'(0)$ | $-751/2\ \nu$ | $-751/2\ \nu$ |
| $S'(0)$ | 1207/28 | 5035/84 |
| $Z''(0)$ | $1207/28 + 6603\ \nu^2$ | $5035/84 + 6603\ \nu^2$ |

$Z''_+(0) - Z''_-(0) = -101/6$

## $Z_+(t) - Z_-(t) = (-101/12)\ t^2 + o(t^2)$

For any fixed positive viscosity, the two smooth local solutions have the same initial Z and Z′, but different Z″. Their enstrophies therefore differ for every sufficiently small positive time. The magnitude inventory plus E, Z, D, H and S cannot determine the entire local future enstrophy curve for every smooth initial field. The different S′ values also obstruct a universal autonomous law for this reduced state.

## Why this is an exact calculation

The full Fourier equation is differentiated twice, retaining every generated mode. Gaussian-rational arithmetic represents all coefficients and Leray denominators exactly. There is no time integration or spectral cutoff. Independent rational rotational-form calculations reproduce the advective-form certificate.

With $B(a,b) = -P[(a \cdot \nabla)b]$, differentiate $u_t = B(u,u) + \nu \Delta u$. Then $Z'' = \|\mathrm{curl}(u_t)\|^2 + \langle \mathrm{curl}(u), \mathrm{curl}(u_{tt}) \rangle$. The complete finite recurrence and modewise rational contributions are in the proof and JSON.

Optional numerical corroboration: 24 independently constructed directional-derivative checks; maximum absolute discrepancy 6.253e-13. This is a check on the certificate, not its mathematical premise.

This does not rule out useful statistical prediction on a restricted distribution, inequalities or bounds, phase-sensitive enlarged states, or models with memory. It says nothing about the specific horizon T=0.5, the forecast quality of the six tested channels, or global regularity. No novelty claim is made.

Sources: phase2/EXACT_OBSTRUCTION.md and results/phase2/exact_obstruction.json. Reproduce: python -m phase2.exact_obstruction. The default report includes this page when that certificate is present; --skip-exact regenerates only the prospective prediction report.